

Automotive Electronics: Shifting Gears

The growing automotive industry and continuous technological developments have transformed our driving experience completely. With new electronic technologies and features in automobiles, the automotive electronics sector in India is poised to be one of the fastest growing markets in the world

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Today cars are equipped with unprecedented number and type of features and gadgets. The replacement of mechanical parts by electronic and mechatronic parts has boosted the automotive electronics industry.

Electronic components in high-end vehicles make up more than 30 per cent of the production cost, and this is expected to rise further. The auto components industry in India is likely to reach a turnover of US \$115 billion by 2020-21 from the existing \$40 billion, and at the same time components export could cross \$30 billion from the current \$10 billion. But these could include non-electronic components as well.

Electronics manufacturing is expected to catch up in India and this could enable the industry to meet dynamic and modern requirements, especially in the automotive domain. Our domestic market is attracting original equipment manufacturers (OEMs) who have a strong need for localisation and the recession-affected global markets are hunting for low-cost vendors to optimise their operations. This could make India one of the fastest growing automotive markets in the world, offering substantial growth opportunities for the various manufacturers.

Also, according to IMS Research, "The global market for automotive electronics is set to rise to \$240 billion in 2020, up more than 50 per cent from \$157 billion in 2010, driven to new levels of prominence by government and automaker safety initiatives." Where is

this growth coming from?

Growth factors

Electronics industry has taken giant leaps in the last few years and, at the same time, it has contributed to the growth of automotive industry as well. Importantly, the main factor for the growth of automotive electronics in Indian market has been the technology



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absorption rate, which has increased manifolds in the recent years.

When we look at our current scenario, the technology absorption rate is very high as compared to earlier times. In terms of various technologies available, we are almost at par with the rest of the world. This fast absorption of technologies has resulted in the simplification of designing processes, reduction in manufacturing costs and at the same time increasing features and safety in automobiles. Thus, electronics in automobiles is now the main focus amongst automakers as well as consumers.

The following are some of the key areas which automotive brands state to be the biggest growth factors of electronics in automotive.

Reduced costs. Properly designed and implemented, electronics can reduce cost while increasing flexibility and reliability. Electronics can define its role in reducing production cost, increasing product quality, meeting tighter standards and customising vehicle features, all within shorter business cycles, which is driving the increased focus on the role of electronics within the automakers.

Selling features. The smartphone era has created a huge automation and assistance demand amongst the consumers in almost every sphere, and automotive is no exception. These electronics innovations have become an essential selling feature for the automobiles. Technologically advanced engine management, safety systems, infotainment and driving assistance are the main drivers of growth for the automotive electronics market.

Safety. Safety requirement is another factor which has forced automotive industry to incorporate electronics in automobiles. According to the United Nations (UN), worldwide 1.3 million people around the world die in traffic accidents each year, and 50 million are injured. The UN has thus announced 2011-20 as the 'Decade of Action for Road Safety' with the objective to reduce the forecast number of road deaths by 50 per cent.

One of the five pillars that the UN has described for this project is 'safer vehicles.' This has driven the industry to focus on the crash avoidance tech-